Sagar Parekh

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About Me

Robotics researcher with a track record spanning foundational and applied research, specializing in robot learning across Imitation Learning, Reinforcement Learning, generative models, deployment on fixed and mobile platforms, and adaptive policy design. Driven by a deep focus on generalization and adaptivity for robot policies. Experienced in developing hardware and software stacks for robots, as well as using physics-based simulators for training and validation. Proven collaborator with 10+ researchers across diverse multi-disciplinary projects. Experience mentoring both undergraduate and graduate students in robotics by organizing outreach programs.

Professional Experience

Graduate Research Assistant 2021 – 2026

Collab, Virginia Tech

- Designed a state compression technique that boosts out-of-distribution *generalization of Imitation Learning policies* under environmental perturbations, improving robustness to unseen conditions.
- Built a *video-based Imitation Learning* pipeline capable of generalizing to long-horizon manipulation tasks from a single demonstration through semantic scene breakdown.
- Leveraged VLMs to reason over vision-language data for generalized skill extraction, and LLMs for generating executable code for robotic manipulation.
- Led the development of an adaptive robot control policy combining Representation Learning with Reinforcement Learning, enabling *rapid online co-adaptation* to novel human partners with minimal interaction data.
- Developed a Meta Learning approach for adaptive dataset balancing, *improving sample efficiency and generalization* of Imitation Learning policies across diverse, heterogeneous data sources.
- Deployed a vision-based control system for high-precision manipulation (meat-cutting) on a collaborative robot arm, demonstrating real-world robotic application.
- Implemented a discrete autoencoder to learn explainable, task-relevant behavioral abstractions from low-level sensorimotor data, supporting *interpretable skill discovery.*

Graduate Teaching Assistant 2025 – 2026

Dept. of Mechanical Engineering, Virginia Tech

- Designed and deployed a PyBullet-based physics simulator as a pedagogical tool, enabling students to gain hands-on experience implementing and evaluating robot control policies in simulation.
- Mentored and provided individualized technical guidance to 90+ students in developing and deploying control algorithms on real hardware, deepening their understanding of robot kinematics, dynamics, and closed-loop control.
- Co-instructed a course on visual robot policy learning for a combined cohort of 60 undergraduate and graduate students, delivering lectures and hands-on curriculum spanning imitation learning, perception, and end-to-end policy training.

Research Assistant 2019 – 2021

HCR Lab, IIT Gandhinagar

- Designed and fabricated a custom multi-quadcopter hardware platform, including full sensor suite integration (IMU, encoders, etc.) and mechanical payload attachment mechanisms for cable-suspended transport.
- Engineered onboard sensing and real-time state estimation pipelines, fusing multi-modal sensor data for robust payload state tracking under dynamic flight conditions.
- Developed a human-in-the-loop shared autonomy framework for cooperative multi-UAV payload transport, enabling seamless arbitration between human intent and learned autonomous control policies.

Select Publications

Robot Learning & HRI

"Transmask: Masked state representation through learned transformation." **Sagar Parekh**, Preston Culbertson, and Dylan P. Losey. Autonomous Robots, 2026 (In Review).

"A Pivot-Based Kirigami Utensil for Hand-Held and Robot-Assisted Feeding." Keone Leao, Grace Brotherson, Ian Mischel, **Sagar Parekh**, and Dylan P. Losey. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.

"Towards Balanced Behavior Cloning from Imbalanced Datasets." **Sagar Parekh**, Heramb Nemlekar, and Dylan P. Losey. Autonomous Robots, 2026.

"Using high-level patterns to estimate how humans predict a robot will behave." **Sagar Parekh**, Lauren Bramblett, Nicola Bezzo, Dylan P Losey. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2025.

"Surveying Processor Perceptions of Automation in Meat Processing." Ryan Wright, **Sagar Parekh**, Dylan P. Losey, and Robin White. Applied Food Research, 2025.

"Safe and Transparent Robots for Human-in-the-Loop Meat Processing." **Sagar Parekh**, Casey Grothoff, Ryan Wright, Robin White, and Dylan P. Losey. Scientific Reports, 2026 (In Review)

"A unified framework for robots that influence humans over long-term interaction." Shahabedin Sagheb, **Sagar Parekh**, Ravi Pandya, Ye-Ji Mun, Katherine Driggs-Campbell, Andrea Bajcsy, Dylan P Losey. The International Journal of Robotics Research, 2026 (In Review).

"View: Visual imitation learning with waypoints." Ananth Jonnavittula, **Sagar Parekh**, Dylan P. Losey. Autonomous Robots, 2025.

"Safely and autonomously cutting meat with a collaborative robot arm." Ryan Wright, **Sagar Parekh**, Robin White, Dylan P Losey. Scientific Reports, 2024

"Learning latent representations to co-adapt to humans." **Sagar Parekh**, Dylan P Losey. Autonomous Robots, 2023.

"Rili: Robustly influencing latent intent." **Sagar Parekh**, Soheil Habibian, Dylan P Losey. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2022.

"Learning latent actions without human demonstrations." Shaunak A Mehta, **Sagar Parekh**, Dylan P Losey. International Conference on Robotics and Automation (ICRA), 2022.

Dynamics and Control

"On-board cable attitude measurement and controller for outdoor aerial transportation." Pratik Prajapati, **Sagar Parekh**, Vineet Vashista. Robotica, 2022.

"On the human control of a multiple quadcopters with a cable-suspended payload system." Pratik Prajapati, **Sagar Parekh**, Vineet Vashista. IEEE international conference on robotics and automation (ICRA), 2020.

"Collaborative transportation of cable-suspended payload using two quadcopters with human in the loop." Pratik Prajapati, **Sagar Parekh**, Vineet Vashista. IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), 2019.

Miscellaneous Projects

Adaptive Robot Policy for Enhanced Human-Robot Interaction

- Leveraged large language models (LLMs) as a foundation for designing and finetuning dense reward models to guide robot policy learning.
- Trained adaptive robot policies via reinforcement learning, optimizing for behaviors that generalize across diverse human partners.

Vision-Based Reinforcement Learning for Autonomous Driving

- Designed a convolutional autoencoder for compact, task-relevant feature extraction from raw RGB-D camera observations.
- Implemented a transformer-based Soft Actor-Critic (SAC) policy, enabling end-to-end training from visual inputs to control actions.
- Trained and evaluated policies achieving robust autonomous lane-keeping and collision avoidance in simulation.

Education

Virginia Tech

PhD in Robotics — *2021 – 2026*
Dissertation: From Pixels to Partners: A Hierarchical Approach to Adaptive Human-Robot Collaboration

Nirma University

BS in Mechanical Engineering — *2015 – 2019*
Thesis: Development of a physics based simulator for human-in-the-loop aerial transportation.